Division elimination for low-degree divisors in ordinary and border circuit complexity

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Editorial note. Faithful editable TeX transcription of formal author Theorems18/22 as one division-elimination group, complete Lemma17 construction and ordinary/border continuations, full long-division AppendixE Lemma70 and monic-transformation AppendixC Lemma68 proofs. Required field, circuit, quotient/remainder, border and standard coefficient-extraction/homogenization/PIT context and complete published bibliography retained. Original expressions and assumptions preserved without mathematical correction or additional proofs; related univariate and constant-free forms remain grouped. Only official CC BY4.0 publication used.

Allowed standard prerequisites (editorial).

- Coefficient extraction and homogenization. Exact author Lemmas61/62 credited to [43] and [45] retained.
- Polynomial Identity Lemma. Exact author Lemma64[38,15,51,41] retained.

Author preliminaries

We denote $\mathbf{x} = (x_1, \ldots, x_n)$. $[n]$ denotes $\{1, \ldots, n\}$. For a polynomial $f \in \mathbb{F}[\mathbf{x}]$, we denote up to degree-d part as $\mathrm{Hom}_{\leq d} f$ and $|f|_0$ as the sparsity or the number of monomials in f. For a differentiable function $f(x)$, we denote $f^{(k)}(x) := d^k f/dx^k$, as the k-th derivative of f. We also recall the definition of gcd of two polynomials f, g in the ring $\mathbb{F}[\mathbf{x}]$: $\gcd(f, g) =: h \iff h \mid f, h \mid g$, and $h' \mid f, g \implies h' \mid h$. It is unique up to constant multiples.

Field. We denote the underlying field as $\mathbb{F}$ and assume that it is algebraically closed. All our results hold when the characteristic is large or not algebraically closed, as we can go to polynomial extensions and work with it.

div and mod operations. For polynomials f and $g \in \mathbb{F}[\mathbf{x}]$, if $f = g \cdot h + r$, where $h, r \in \mathbb{F}[\mathbf{x}]$ such that $\deg(r) < \deg(g)$, then h is called the quotient, denoted $f \operatorname{div} g$, and r is called the remainder, denoted $f \bmod g$. Operation mod may not be well-defined in the multivariate settings, however, if one assumes g to be monic in a variable say x_n, it is always well-defined (by thinking g to be a univariate in x_n). A polynomial g is monic in x_n if the leading coefficient (the nonzero coefficient of highest degree) of x_n is a non-zero constant in $\mathbb{F}$. Of course, if $g \mid f$, then $f \operatorname{div} g = h$ and $f \bmod g = 0$, irrespective of monic-ness.

Circuit complexity. For a polynomial $f \in \mathbb{F}[\mathbf{x}]$, we define the complexity $L_{\mathbb{F}}(f)$ of f to be the length of the smallest division-free arithmetic circuit (with only $\{+, -, \times\}$ gates) computing f. We also define the complexity $\tau_{\mathbb{F}}(f)$ of f to be the length of the smallest division and constant-free arithmetic circuit computing f (all the constants are made from 1), for formal definition see Section 5.2 of the original paper. We will remove subscript $\mathbb{F}$ when the underlying field is clear from the context.

Border complexity context (author Section 3.3). The notion of border (equivalently, approximative) complexity is important in computer science. This concept popped up from early works on matrix multiplication and border rank of tensors, see [10]). Whether approximation of polynomials provides any additional computational power is a natural question which fundamentally motivated the foundation of Geometric Complexity theory (GCT). The notion of border complexity can be motivated through two ways: topological and algebraic, and both the perspectives are known to be equivalent [1]. For further details, we refer to [20,36]. In this paper, we only work with algebraic approximation upper bounds. In the algebraic definition, one can talk about the convergence $\epsilon \to 0$. Here, one can see ϵ as a formal variable and $\mathbb{F}(\epsilon)$ as the function field. For an algebraic complexity class $\mathcal{C}$, the approximation is defined as follows [7,Definition2.1].

Definition 20 (Approximative closure of a class [7]). Let $\mathcal{C}$ be an algebraic complexity class over field $\mathbb{F}$. A family (f_n) of polynomials from $\mathbb{F}[\mathbf{x}]$ is in the class $\overline{\mathcal{C}}(\mathbb{F})$ if there are polynomials $f_{n;i}$ and a function $t : \mathbb{N} \mapsto \mathbb{N}$ such that g_n is in the class $\mathcal{C}$ over the field $\mathbb{F}(\epsilon)$ with $g_n(\mathbf{x}) = f_n(\mathbf{x}) + \epsilon f_{n,1}(\mathbf{x}) + \epsilon^2 f_{n,2}(\mathbf{x}) + \cdots + \epsilon^{t(n)} f_{n,t(n)}(\mathbf{x})$.

Definition 21 ([8,Defn.3.1]). Let $f \in \mathbb{F}[\mathbf{x}]$. The border complexity $\underline{L}(f)$ is the smallest number r, such that there exists F in $\mathbb{F}(\epsilon)[\mathbf{x}]$ satisfying $F|_{\epsilon=0} = f$ and $L_{\mathbb{F}(\epsilon)}(F) \leq r$. Note that the circuit of F may be using $1/\epsilon$ in an intermediate step. So, we cannot merely assign $\epsilon = 0$ and get a ϵ-free circuit. Also, the ϵ-degree can be exponential in its size (and thus cannot be interpolated), see [8,Theorem5.7]). Thus, potentially $\underline{L}(f)$ can be significantly smaller than $L(f)$. The above definition can be used to define closures of complexity class, e.g., $\overline{\mathrm{VP}}$. In this case, one can assume wlog that the degrees of g_n and $f_{n,i}$ are poly(n). It is known to be closed under factoring [8,Theorem4.1]. However, the usual method of Hensel-lifting does not work when the given circuit class computes polynomials of super-polynomial degree. Also, Strassen's method would have

a dependency on the degree of the final polynomial. However, we can prove Theorem 18 analogously, in the border sense.

Lemma 61 (Coefficient-Extraction). Coefficient-extraction in arithmetic circuits is easy using interpolation, see the folklore lemma below, for a proof see [43]. Let $L(f) = s$ with $f \in \mathbb{F}[\mathbf{x}]$ and $f = \sum_{0 \leq i \leq d} f_i x_n^i$ with $f_i \in \mathbb{F}[x_1, \ldots, x_{n-1}]$. Then there is a circuit C of size $O(sd^2)$ computing $f_0, f_1, \ldots, f_d$.

Lemma 62 (Homogenization). The next lemma is a homogenization trick, used in [45]. For a proof, see [43,Theorem2.2]. If f has an arithmetic circuit of size s, then for any d, there is a circuit of size $O(sd^2)$ computing $\mathrm{Hom}_{\leq d} f$.

Lemma 64 (Polynomial Identity Lemma). Let $p(\mathbf{x})$ be an n-variate nonzero polynomial of degree d. Let $S \subseteq \mathbb{F}$ be a finite set. Then $\Pr_{\alpha \sim S^n}[p(\alpha) = 0] \leq d/|S|$. Here, $\alpha \in S^n$ is picked independently and uniformly at random. This is the standard Polynomial Identity Lemma [38,15,51,41].

Formal author division-elimination main results

Theorem 18 (Division elimination for low-degree divisor). Let the polynomials $f, g, h \in \mathbb{F}[\mathbf{x}]$ such that $f = g/h$, with $L(g) = s_1$, $L(h) = s_2$, and $\deg(h) = d$. Then, $L(f) \leq O((s_1 + s_2)d^2)$.

Theorem 22 (Division elimination in border complexity). Let $f, g, h \in \mathbb{F}[\mathbf{x}]$, such that $f = g/h$, with $\underline{L}(g) = s_1$, $\underline{L}(h) = s_2$, and $\deg(h) = d$. Then, $\underline{L}(f) \leq O(s_1 d^2 + s_2 d^4)$.

Complete author low-degree division tools

Theorem 13. Let g, h be polynomials in $\mathbb{F}[x]$. If $L(g) = s$ and $\deg(h) = d$, then both $L(g \operatorname{div} h)$ and $L(g \bmod h)$ have complexity $O(sd)$.

Proof. Suppose C is a circuit of size s which computes g. We split every gate Φ in C into two gates Φ_1 and Φ_2, to make a new circuit C', which computes both $g \operatorname{div} h$ and $g \bmod h$. If Φ is computing some polynomial ϕ in C, then Φ_1 computes the polynomial $\phi \bmod h$ and Φ_2 computes the polynomial $\phi \operatorname{div} h$. The proof is inductive and traverses from bottom to the top. The base case is trivial. At some step, say that we are at a gate Φ. The children gate of ϕ are computing polynomials α and β. Let $\alpha = q_1 h + r_1$, $\beta = q_2 h + r_2$ and $\phi = qh + r$, where the degrees of r, r_1, r_2 are smaller than d. So in the new circuit C', we have already computed r_1, q_1, r_2, q_2. If Φ is a $\pm$ gate then it is clear that $r = r_1 \pm r_2$ and $q = q_1 \pm q_2$. If Φ is a $\times$ gate then we have:

$$r = (r_1 r_2) \bmod h,$$
$$q = q_1 q_2 h + r_1 q_2 + r_2 q_1 + (r_1 r_2) \operatorname{div} h.$$

We know that r is a polynomial of degree at most $d - 1$. Since $\deg(r_1 r_2) \leq 2d - 2$, we get that $\deg((r_1 r_2) \operatorname{div} h) \leq d - 2$. Therefore, we trivially have that:

$L(r) = O(d)$ and $L((r_1 r_2) \operatorname{div} h)) = O(d)$. Hence we can compute r, q using additional $O(d)$ many gates. Thus, C' has at most $O(sd)$ many gates. Hence $L(g \operatorname{div} h) = O(sd)$ (same for $g \bmod h$). □

Corollary 14. For $f, g, h \in \mathbb{F}[x]$, if $f = g/h$ with $L(g) = s$ and $\deg(h) = d$ then $L(f) = O(sd)$.

Remark 15. The polynomial $f_d := 1 + \cdots + x^d = (x^{d+1} - 1)/(x - 1)$ has $O(\log d)$ size circuit. This can also be shown via a recursive computation argument. Can we expect both div and mod to have $\text{poly}(s, \log d)$-size circuits? We show that it is highly unlikely unless factoring is easy, see Theorem 54 in the original paper for details.

Lemma 16 (same as Appendix E Lemma 70). Suppose $g = \sum_{i \le d_1} g_i x^i$ and $h = x^{d_2} + \sum_{i < d_2} h_i x^i$, in $\mathbb{F}[x]$. Suppose $g = hq + r$, with $r = \sum_{i < d_2} r_i x^i$ and $q = \sum_{i \le d_1 - d_2} q_i x^i$. Then, there is a circuit of size $O(d_1 d_2)$, whose inputs are all h_i, g_i and outputs are all r_i, q_i.

Complete Appendix E proof (Lemma 70). We shall denote the desired circuit by C_{d_1,d_2}. So we want:

$$C_{d_1,d_2}(g_0, g_1, \ldots, g_{d_1}, h_0, h_1, \ldots, h_{d_2}) \\ = (r_1, r_2, \ldots, r_{d_2-1}, q_0, q_1, \ldots, q_{d_1-d_2}).$$

If $d_1 < d_2$, we know that $q = 0$. Hence:

$$C_{d_2-1,d_2}(g_0, g_1, \ldots, g_{d_1-1}, h_0, h_1, \ldots, h_{d_1}) = (g_1, g_2, \ldots, g_{d_1-1}).$$

If $d_1 > d_2$, we perform a long division step:

$$g \leftarrow g - h x^{d_1-d_2} g_{d_1} \\ = \sum_{i \le d_1 - d_2 - 1} g_i x^i + \sum_{i \ge d_1 - d_2}^{d_1 - 1} (g_i - h_{i-(d_1-d_2)} g_{d_1}) x^i.$$

Note that, we can set $q_{d_1-d_2} = g_{d_1}$. Define:

$$\mathbf{g} \stackrel{\text{def}}{=} (g_0, g_1, \ldots, g_{d_1-d_2-1}, g_{d_1-d_2} - h_0 g_{d_1}, \ldots, g_{d_1-1} - h_{d_2-1} g_{d_1}).$$

Then we have:

$$C_{d_1,d_2}(g_0, g_1, \ldots, g_{d_1}, h_0, h_1, \ldots, h_{d_2}) \\ = (C_{d_1-1,d_2}(\mathbf{g}, h_0, h_1, \ldots, h_{d_2}), g_{d_1}). \quad (8)$$

Hence if $S(d_1, d_2)$ is the size of C_{d_1,d_2} then Equation (8) implies that $S(d_1, d_2) = S(d_1 - 1, d_2) + 2d_2$ and $S(d_2 - 1, d_2) = 2d_2 - 1$. Therefore $S(d_1, d_2) \le 2d_1 d_2$. □

Complete author monic transformation (Appendix C)

Given any polynomial $p(\mathbf{x})$ in variables $\mathbf{x} = (x_1, \ldots, x_n)$, there is a standard trick to make it monic in x_n by applying a linear transformation on the variables: for $\alpha = (\alpha_1, \ldots, \alpha_{n-1}) \in \mathbb{F}^{n-1}$, let $\tau_\alpha : x_i \mapsto \alpha_i x_n + x_i$, for $i \in [n-1]$, and $x_n \mapsto x_n$. Note that $\deg(\tau_\alpha(p)) \leq \deg(p)$ [it may decrease because of non-trivial cancellations]. It is easy to see that τ_α is an invertible map. We show that $\tau_\alpha(p)$ is monic in x_n, for a random transformation τ_α i.e. when $\alpha \in \mathbb{F}^{n-1}$ chosen randomly. In fact, we show that this map can simultaneously make polynomials monic given that the field $\mathbb{F}$ is sufficiently large.

Lemma 68 (Monic Transformation). Let $p_1(\mathbf{x}), \ldots, p_m(\mathbf{x})$ be m-many polynomial of degree d. Let $S \subseteq \mathbb{F}$ be a finite set. For $\alpha \in S^{n-1}$, picked independently and uniformly at random,

$$\Pr\left[\bigwedge_{i=1}^{m} \tau_\alpha(p_i(\mathbf{x})) \text{ is monic in } x_n\right] \geq 1 - \frac{dm}{|S|}.$$

Proof. Consider the terms of degree d of a non-zero polynomial $p \in \mathbb{F}[\mathbf{x}]$. Define the set

$$T := \{\beta = (\beta_1, \ldots, \beta_n) : |\beta|_0 = \sum_i \beta_i = d, \text{ and } \operatorname{coef}_{\mathbf{x}^\beta}(p) \neq 0\}.$$

We also denote $\beta' = (\beta_1, \ldots, \beta_{n-1})$, the first $n-1$-coordinates of β, and similarly $\mathbf{x}' = (x_1, \ldots, x_{n-1})$. Note that $\tau_\alpha(\mathbf{x}^\beta) = \alpha^{\beta'} x_n^d + (\text{lower terms in } x_n)$. Observe that the homogeneous component of degree d in $\tau_\alpha(p)$ can be written as $a_{d,p}(\mathbf{x}) = \sum_{\beta \in T} c_\beta \tau_\alpha(\mathbf{x}^\beta)$, for some constants c_β. Trivially, $a_{d,p}$ is a nonzero polynomial, and moreover,

$$a_{d,p}(\alpha) = \left(\sum_{\beta \in T} c_\beta \alpha^{\beta'}\right) x_n^d + (\text{lower terms in } x_n).$$

In order to make $\tau_\alpha(p)$ monic in x_n, we want $(\sum_{\beta \in T} c_\beta \alpha^{\beta'}) \neq 0$. So, define, another polynomial $b_{d,p}(\mathbf{x}') = (\sum_{\beta \in T} c_\beta \mathbf{x}'^{\beta'})$. It can have degree atmost d. As we want each $\tau_\alpha(p)$ monic where $p = p_m(\mathbf{x})$, it suffices to find α such that $\prod_{i \in [m]} b_{d,p_i}(\alpha) \neq 0$. Note that $\deg(\prod_{i \in [m]} b_{d,p_i}(\mathbf{x})) \leq d \cdot m$. Thus, when we pick α at random, the probability that $\prod_{i \in [m]} b_{d,p_i}(\alpha) = 0$, is at most $\leq dm/|S|$, from Lemma 64. Hence, the conclusion follows. □

Complete author Section 3.2 main construction

Now we prove the following Lemma 17 which shows that both div and mod have low complexity when the divisor has low-degree and monic (in fact, constant leading-coefficient suffices).

Lemma 17 (Main Lemma). Let the polynomials $g, h \in \mathbb{F}[\mathbf{x}]$ such that h is monic in x_n, $L(g) = s_1$, $L(h) = s_2$, and $\deg_{x_n}(h) = d$. Then, both $L(g \operatorname{div} h), L(g \bmod h) \leq O((s_1 + s_2)d^2)$.

Proof. Suppose C is a circuit of size s_2 which computes h and C_g is the circuit of size s_1 which computes g. By using Lemma 61, there is a circuit of size $O(s_2 d^2)$, which computes $h_0, \ldots, h_{d-1}$. Now, we split every gate F in C into $d+1$ gates $F_0, F_1, \ldots, F_d$. Suppose, the gate F is computing a polynomial P_F. Let $P_F \bmod h = \sum_{i<d} p_i x_n^i$. Then we want the property that F_i computes p_i for $i < d$. And if $i = d$ then F_i computes $P_F \operatorname{div} h$. Suppose F is a $+$ gate in C with children gates computing the polynomials a and b. Again express $a \bmod h = \sum_{i<d} a_i x_n^i$ and $b \bmod h = \sum_{i<d} b_i x_n^i$. It is clear that

$$(a + b) \bmod h = a \bmod h + b \bmod h.$$

Therefore $p_i = a_i + b_i$. It is also clear that $P_F \operatorname{div} h = a \operatorname{div} h + b \operatorname{div} h$. Suppose F is a $\times$ gate in C with children gates computing the polynomials a and b. Again express $a \bmod h = \sum_{i<d} a_i x_n^i$ and $b \bmod h = \sum_{i<d} b_i x_n^i$. It is clear that:

$$(a \cdot b) \bmod h = (a \bmod h \cdot b \bmod h) \bmod h.$$

For div, we have that:

$$\begin{aligned} P_F \operatorname{div} h = {} & a \operatorname{div} h \cdot b \operatorname{div} h \cdot h + b \operatorname{div} h \cdot a \bmod h \\ & + a \operatorname{div} h \cdot b \bmod h + (a \bmod h \cdot b \bmod h) \operatorname{div} h. \end{aligned}$$

We have already computed $a \bmod h, b \bmod h, a \operatorname{div} h, b \operatorname{div} h$. So, we only need to compute $(a \bmod h \cdot b \bmod h) \bmod h$ and $(a \bmod h \cdot b \bmod h) \operatorname{div} h$. Since we have already computed a_i, b_i for all $i < d$, by using Lemma 16, we can compute all the p_i and $(a \bmod h \cdot b \bmod h) \operatorname{div} h$ in $O((2d-2)d) = O(d^2)$ many gates. Therefore the new circuit has $O(s_1 d^2)$ has many gates. Also we used $O(s_2 d^2)$ gates to computes $h_0, \ldots, h_{d-1}$. Hence,

$$L(g \operatorname{div} h) = O((s_1 + s_2)d^2), \qquad L(g \bmod h) = O((s_1 + s_2)d^2).$$

□

The following theorem settles Problem 1, when the divisor has small degree (proving Theorem 4).

Proof of Theorem 18. The above Lemma 17 shows that when h is monic in x_n, the upper bound holds. Let $\tau : \mathbb{F}[\mathbf{x}] \longrightarrow \mathbb{F}[\mathbf{x}]$, be an invertible monic transformation (sends $x_i \mapsto \alpha_i \cdot x_n + x_i$, where $\alpha_i \in \mathbb{F}$) s.t. $\tau(h)$ is monic wrt x_n, such transformation exists (Lemma 68). Note that, $L(\tau(g)) \leq s + n = O(s_1)$, and $L(\tau(h)) \leq s_2 + n = O(s_2)$. Moreover, as τ is degree-preserving, $\deg_{x_n}(\tau(h)) = d$. So, apply Lemma 17 to conclude that $\tau(f) = \tau(g) \operatorname{div} \tau(h)$, has a circuit of size $O((s_1 + s_2)d^2)$. We apply τ^{-1} again (which is just a additive n-blowup) to finally deduce that

$$L(f) \leq O((s_1 + s_2)d^2).$$

□

Remark 19. This proof holds when one replaces L by τ, i.e. the constant-free circuit complexity (for definition, see Section 5.2). Note that, neither div nor mod introduce any new constant in the process. Moreover, one can choose the α_i to be explicit and poly($\log d$)-computable so that τ is a monic invertible map. This establishes the claim.

Complete author Section 3.3 border proof

Proof of Theorem 22. By definition, there exists $G, H \in \mathbb{F}(\epsilon)[\mathbf{x}]$, of size at most s_1 and s_2, respectively, such that $G := g+\epsilon\cdot\widetilde{g}(\mathbf{x},\epsilon)$, and $H := h+\epsilon\cdot\overline{h}(\mathbf{x},\epsilon)$, where $\widetilde{g}, \overline{h} \in \mathbb{F}[\epsilon, \mathbf{x}]$. We note that, $\deg_{\mathbf{x}}(H)$ can be larger than d. However, using Lemma 62, we know that $L_{\mathbb{F}(\epsilon)}(\mathrm{Hom}_{\leq d}H) \leq O(s_2d^2) =: s_2'$. We denote $\widetilde{H} := \mathrm{Hom}_{\leq d}H$. It is important to observe that $\widetilde{H}|_{\epsilon=0} = h$. By definition, there exists m (could be $\exp(s_2')$) such that

$$\widetilde{H} := h + \epsilon\cdot\widetilde{h}(\mathbf{x},\epsilon) = h + \sum_{j\in[m]} \epsilon^j h_j(\mathbf{x}), \qquad h_j \in \mathbb{F}[\mathbf{x}].$$

Let $\tau : \mathbb{F}[\mathbf{x}] \longrightarrow \mathbb{F}[\mathbf{x}]$, be an invertible monic transformation (sends $x_i \mapsto \alpha_i \cdot x_n + x_i$, where $\alpha_i \in \mathbb{F}$) s.t. $\tau(h)$ and each $\tau(h_j)$, for $j \in [m]$ is monic wrt x_n; such transformation exists (Lemma 68). Note that, $L_{\mathbb{F}(\epsilon)}(\tau(G)) \leq O(s_1)$ and $L_{\mathbb{F}(\epsilon)}(\tau(\widetilde{H})) \leq O(s_2')$. Further, $\deg_{\mathbf{x}}(\tau(\widetilde{H})) = d$, as τ is a degree-preserving map. We also have the following identities:

$$\tau(\widetilde{H}) = \tau(h) + \epsilon\cdot\tau(\widetilde{h}), \qquad \tau(G) = \tau(f)\cdot\tau(h) + \epsilon\cdot\tau(\widetilde{g}).$$

By assumption, the leading coefficient of x_n in $\tau(\widetilde{H})$ (call it α) is in $\mathbb{F}[\epsilon]$ (in fact, $\alpha \not\equiv 0 \bmod \epsilon$). This basically makes $\tau(\widetilde{H})$ a monic polynomial over $\mathbb{F}(\epsilon)[\mathbf{x}]$. Therefore, $\mathrm{div}\,\tau(\widetilde{H})$ and $\mathrm{mod}\,\tau(\widetilde{H})$ now make sense over $\mathbb{F}(\epsilon)[\mathbf{x}]$. By simple division, we have

$$\tau(G)\,\mathrm{div}\,\tau(\widetilde{H}) = \tau(f) + \epsilon\cdot(\tau(\widetilde{g}) - \tau(f)\cdot\tau(\widetilde{h}))\,\mathrm{div}\,\tau(\widetilde{H}). \quad (1)$$

Note that, Lemma 17 implies $L_{\mathbb{F}(\epsilon)}(\tau(G)\ \mathrm{div}\ \tau(\widetilde{H})) = O((s_1 + s_2')d^2)$. By definition of $\underline{L}$ and Equation (1), it is trivial to conclude that $\underline{L}(\tau(f)) \leq O((s_1 + s_2')d^2) = O(s_1d^2 + s_2d^4)$. As τ is invertible, we can get back f by applying τ^{-1} (incurring n-additive blowup). This finally shows

$$\underline{L}(f) \leq O(s_1d^2 + s_2d^4).$$

□

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